SensAI: Technical Report

Akachi David Nwanze · BSc Software Engineering (ML) · African Leadership University · Supervisor: Elvira Immaculate Khwatenge · October 2026 · Repository: github.com/S1rDavid9/SensAI_capstone · Video: youtu.be/7H3JrJ6SkBk

1. What SensAI is

SensAI is a web app for learners aged 8 to 14 that adapts short maths and pattern puzzles to how a learner is working. After every answer it runs a closed loop: **Stage 1**, a classifier, estimates a behavioural state from the last five answers (on task, rushing/under-challenged, or struggling/needs support); **Stage 2**, a Deep Q-Network (DQN) trained in simulation, chooses one of eight scaffolding actions; guardrails rule out actions no teacher would choose; and the app applies the action by choosing the next question and showing a short message or hint. Engagement is treated strictly as observable behaviour (Fredricks et al., 2004): the system does not claim to detect emotions. This milestone delivers the simulation and policy (Phase 1), the data-collection pilot with 31 learners at Excella School, Kigali, the Stage 1 classifier trained on that data, and the closed-loop adaptive app as the final MVP.

2. Scope alignment and feature validation

Features are mapped to the approved proposal **as corrected by my supervisor's feedback**; where that feedback changed the plan, the changed version is what was built, and the deviation is stated.

Planned feature	Intended purpose	Status	Evidence and notes
Telemetry logging engine (Objective 1)	Capture response time, errors, hints, time on task without cameras or sensors	**Completed**	Learner screen logs answers, time to first input, edits, 20-s pauses, leaving the window, check-ins, and now device/browser (`pilot/services/learning.py`, `pilot/web/src/screens/Learner.tsx`)
Simulation and DQN, benchmarked against PPO, A2C, REINFORCE (Objective 2)	Learn a scaffolding policy before real data exists	**Completed, extended**	Done in Phase 1, so it is reported as a result, not future work (supervisor feedback). Extended with **simulator v2** after the pilot exposed two reward loopholes (`environment/custom_env.py`, decision 12; `logs/v2/evaluation.md`)
Operational behavioural states (feedback)	Define engagement-risk states by observable behaviour	**Completed**	`BehaviouralState`: on task, rushing, struggling, stuck, each mapped to a DQN state slot (`pilot/core/states.py`); the class diagram's affective labels are replaced
Ground-truth labelling (Objective 3)	Validate telemetry against independent labels	**Partial**	Learner self-report check-ins: 94 collected. Two-observer momentary time sampling built and used, but only 24 codes and no paired codes, so inter-rater agreement (Cohen, 1960) could not be computed
Feasibility pilot, 10–20 learners (Objective 4; setting named per feedback)	Collect real telemetry from school-age learners	**Completed with deviation**	Excella School, Kigali (private), IREC approval, guardian consent and learner assent; 31 learners started and 30 finished, above the planned 10–20. Run in batches of about 12 on school Chromebooks plus iPhones, an Android phone, iPads, and Mac and Windows laptops, in Chrome and Safari
Stage 1 classifier: logistic regression / shallow XGBoost	Predict the state from telemetry	**Completed**	Leave-one-learner-out cross-validation (CV split by learner, per feedback); `analysis/stage1.py`, `models/stage1/`
Shadow mode (feedback)	Guarantee the DQN did not affect pilot learners	**Completed**	Decisions stored server-side only; verified by `test_shadow_mode_never_reaches_the_learner`
Guardrails (approved in office hour)	Block actions no teacher would choose	**Completed (new)**	`Shield` with rule classes (`pilot/policy/shield.py`)
Integrated web prototype (Objective 4)	Run the full telemetry → state → action loop	**Completed**	Adaptive mode (`pilot/services/modes.py`, `pilot/adaptation/`); demonstrated by the researcher, not used with children
Decision plausibility (defined per feedback)	Raters score DQN decisions 1–3; weighted kappa between two raters	**Completed (one rater)**	78 blind items from 60 real moments, rated by the researcher (`analysis/rating_sheet.py`, `analysis/results/plausibility.json`). The class teacher's ratings, needed for the agreement statistic, are the next step

Data layer: four tables (text) vs five / six (diagrams)	Store sessions, telemetry, labels, decisions	**Completed, reconciled**	Six tables: sessions, answers, events, check-ins, observations, decisions (`pilot/storage/`). SQLite, the proposal's option for a small pilot
Docker deployment (proposal 3.4)	Reproducible deployment	**Not implemented**	Replaced by a locked `uv` environment and a local-network server: the school network could not be relied on, and no cloud service may receive learner data under the ethics approval

3. Technical implementation

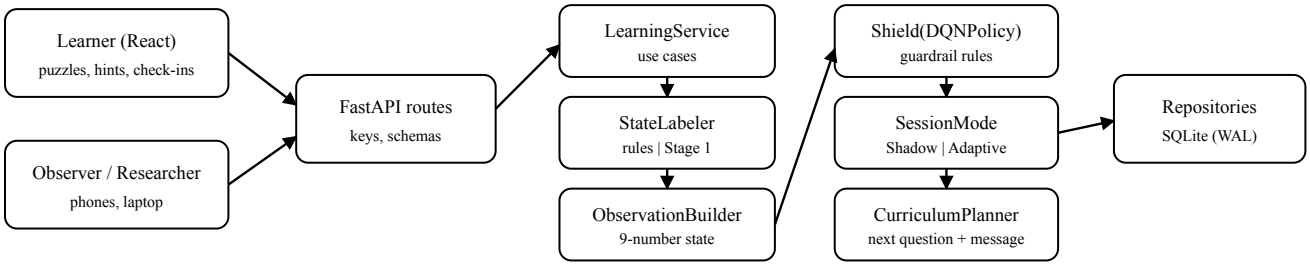


Figure 1. Request path for one answer. Each box is a class or module; arrows show dependencies (top-level wiring is in `create_app`).

Backend/frontend interaction. The React (TypeScript) screens talk to a FastAPI backend over JSON. The learner screen sends an event when a question is shown, at the first keystroke, every 20 seconds without input, and on leaving the window, then posts the answer with client-measured timings. The server grades it, stores it, runs Stage 1 → Stage 2 and, in adaptive mode, stores the next question and the support message, which the next `/question` call returns. Devices synchronise clocks with `/api/time` (offset taken from the lowest round-trip of three samples) so observations from phones line up with laptop events.

Algorithms and problem-solving

- **DQN policy** (Mnih et al., 2015): a 9 → 64 → 64 → 8 Q-network with experience replay and a target network, trained on 20,000 simulated steps. The pilot revealed two reward loopholes: alternating *advance topic* (bonus) with *decrease difficulty*, and moving on when a learner was confused. Simulator v2 withholds the topic bonus in both cases and makes advancing while confused more likely to keep the learner confused. Result: "confused → move on" fell from 69% to 0% of decisions on held-out seeds; v1 still reproduces exactly (8.50 mean reward). PPO, A2C and REINFORCE collapsed to one or two actions under v2; with five times the training budget PPO still chose one action 84% of the time, so DQN is the policy carried forward.
- **Why reinforcement learning, not teacher rules?** On 200 held-out simulated learners (same learners for every policy, simulator v2), two rule policies written from teaching practice and not tuned to the simulator did far worse than the DQN with guardrails: a fixed rule per state (mean reward 0.43, 60% dropout) and an errors-only difficulty staircase (0.24, 66% dropout), against 7.02 and 10% dropout (paired Wilcoxon $p < 10^{-25}$). Both rule policies kept raising difficulty for learners who were coping until they disengaged; the DQN learned to hold difficulty steady. This holds inside the simulator the DQN was trained in, so it shows the policy learned something non-trivial, not that it would beat a teacher's rules with real children (`analysis/baseline_comparison.py`).
- **Shielding** (Alshiekh et al., 2018): `Shield.decide` masks forbidden actions and takes the arg-max of the remaining Q-values. Both the raw and the shielded action are stored, so results can be reported with and without guardrails. In simulator v2 the guardrails cut dropout from 18% to 10% with no significant change in reward (+0.35, $p = 0.26$).
- **Stage 1**: features from a sliding window of the last five answers; standardised logistic regression with balanced class weights. Error rate and response time follow prior work on response-time effort and disengagement (Beck, 2005; Wise & Kong, 2005) and on gaming behaviour (Baker et al., 2004).
- **Curriculum planner**: each action maps to a target level and a question-type rule; the planner searches levels outward from the target (target, ±1, ±2), prefers the requested type, never repeats a question, and rewrites the message when the bank cannot move as intended (for example "easier" at level 1).

4. Testing, validation and performance

Approach	Why chosen	How conducted and results	Improvements made afterwards

Unit and integration (pytest)	Each class has one job, so it can be tested alone; the API must behave correctly end to end in both modes	77 tests; 96% of `pilot/` statements covered. Cover the question bank, both labellers, every guardrail rule, the planner, database upgrades (including six processes starting at once), consent rules, access keys, invalid inputs (wrong code, wrong question, negative time, unknown event, early finish → 400/403/404/409/422) and full sessions in both modes	Planner message corrected when difficulty cannot change; shadow-mode test added at the supervisor's request
Stress / performance	A class answers at once on one laptop server; failure would lose children's data	`tests/performance/load_test.py` runs the real server and 10, 30, 60 and 100 learners answering all 15 questions with no thinking time. **0 failed requests out of 40,000**. At 30 learners, median 0.04 s and 95th percentile 0.45 s; at 100, 95th percentile about 1.1–1.3 s. Memory 0.2–0.4 GB (one worker)	"database is locked" when several workers started → file-locked initialisation; 2-s first-answer delay → models load at startup; SQLite WAL mode; PyTorch on one thread
Compatibility (Playwright)	Learners used Chromebooks, phones, iPads and laptops	Full adaptive session per target in Chrome 155 (Android phone, Chromebook 1366×768, desktop), WebKit 27.2, the Safari engine (iPhone 13, iPad, Mac) and Firefox 157, plus a low-end phone emulated with a CPU six times slower: **8/8 pass**, no JavaScript errors, no horizontal overflow. Every target renders at 60 frames per second; on the low-end phone page load doubles (115 ms vs 53 ms) while answer-to-feedback time barely changes (299 ms vs 276 ms end to end)	Smallest buttons measured 43 px, under the 44 px touch-target guideline → all learner buttons and inputs at least 48 px
Field / usability (pilot)	Real use by the target age group	31 learners, batches of about 12, on Chromebooks, iPhones (Safari and Chrome, both WebKit, matching the WebKit tests), an Android phone, iPads, and Mac and Windows laptops; 30 of 31 finished; only 4 ever left the window; 94 check-ins answered	Device, browser and screen size are now recorded per session; smaller batches recommended for observation

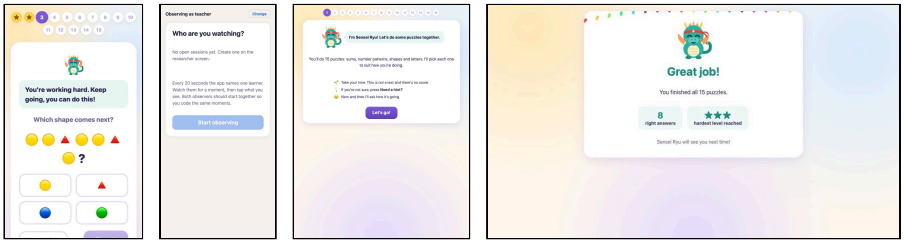


Figure 2. Screenshots from the compatibility run: adaptive support on an iPhone (WebKit), the observer screen on an Android phone (Chrome), the welcome screen on an iPad, and the finish screen on a Chromebook-sized window.

Validation results on the pilot data

Signals vs labels. Error rate in the five questions before a check-in separated "hard/lost" from other answers by 1.46 standard deviations within the same block ($p < 0.001$). Response time alone did not, because the two struggling answers behaved oppositely: "hard" was slow (about 69 s per answer), "lost" was fast (about 23 s) with 74% errors, consistent with rapid guessing (Wise & Kong, 2005). **Stage 1** (leave-one-learner-out, 92 labelled blocks): balanced accuracy 0.67 against 0.33 for the majority baseline; AUC 0.90 for "needs support". Adding more features lowered accuracy (all seven: 0.58; shallow XGBoost: 0.47), so the deployed model uses error rate and response time only. **Stage 1 vs adult observation:** matching each of the 24 observer codes to the question the learner was working on, Stage 1 agreed in only 5 (21%); observers saw struggling or stuck 15 times, and Stage 1 mostly said on task or rushing. Self-report and adult observation appear to capture different things, and Stage 1 judges a five-answer window while an observation is a single moment; with 24 codes from 12 learners this is a warning sign rather than a firm result, and it is the main validity question for the next phase. **Stage 1 → DQN on the 463 real answers**: under-challenged → move on (76%) or challenge (20%); needs support → simplified repeat (52%), easier (29%), encourage (19%); on task → the DQN chose "easier" 70% of the time, which the guardrail blocked (21.8% of all decisions shielded). On-task behaviour is the policy's

known weak spot. **Decision plausibility** (60 real moments, rated blind by the researcher): 85% of the DQN's decisions were rated acceptable or better (mean 2.52 of 3): 100% for under-challenged learners, 95% for learners needing support, but only 60% for learners on task. All 18 guardrail interventions were "make it easier" for an on-task learner; the replacement (usually a hint) was rated the same in 12, better in 2 and worse in 4, so with guardrails the rate was 82%. The guardrail blocks the bad choice, but the policy's next-best action is not a good fallback for a coping learner; a better on-task fallback is the next fix. One rater, who also built the system, and 60 moments: a second rater is needed before relying on these figures.

5. Code quality and OOP practice

- **Layered architecture and separation of concerns:** `core` (domain), `labelling`, `policy`, `adaptation`, `storage`, `services`, `api`. Routes contain no logic; services raise domain errors (`NotFound`, `Conflict`…) that one handler maps to HTTP.
- **Abstraction and polymorphism:** `StateLabeler` (implemented by `RuleBasedLabeler` and `Stage1Labeler`), `ActionScorer` (`DQNPolicy`, and a fake scorer in tests), `GuardrailRule` (`StateRule`, `DifficultyBoundsRule`) and `SessionMode` (`ShadowMode`, `AdaptiveMode`: the Strategy pattern). Inheritance is used only for these interfaces.
- **Encapsulation and SOLID:** one repository per table hides SQL; a `Database` class owns connections and upgrades. New rules or labellers are added without changing existing classes (open/closed); services receive their collaborators in `create_app` (dependency inversion), which is what makes them testable with fakes.
- **Readability and maintainability:** consistent names (`BehaviouralState`, `CurriculumPlanner`), frozen dataclasses for value objects, module docstrings explaining why, settings read once from the environment, and a commit history describing each change.

6. Ethics, limitations and next steps

The pilot ran under ALU Research Ethics Committee approval with guardian consent and learner assent; learners are known only by codes, and no cloud service receives their data. **During the pilot the DQN never affected learners:** everyone saw the same 15 questions and the policy's decisions were only stored. Adaptive mode is demonstrated by the researcher; using it with children needs a new approval. Limitations: one school and one session per learner; only 11 "hard"/"lost" check-ins; observations too sparse for inter-rater agreement, and Stage 1 agreed with observers in only 5 of 24 moments; more learners than the approved 10–20 (a deviation from the approved protocol); plausibility rated by one rater only; and REINFORCE results are not reproducible. Next: a second plausibility rating by the class teacher (for weighted kappa), a better on-task fallback in the guardrails, run a smaller observation session for paired codes, and seek approval to evaluate adaptive mode with learners.

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